

911 Emergency Analytics and Spatial Intelligence System

1. Introduction

Objective

The objective of this project is to analyze historical 911 emergency call records to identify temporal trends, geographic hotspots, and unusual emergency activity patterns using Python-based data analysis and machine learning techniques.

Dataset

- Source: Calls Dataset
- Shape: 86576 rows $\times$ 9 columns
- Features: lat, lng, zip, title, timeStamp, twp, addr, Reason

2. Technologies Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn
- Plotly

3. Data Preprocessing

Describe:

- Missing value handling
- Timestamp conversion
- Feature extraction

Features created:

- Hour
- Month
- DayOfWeek
- Reason

Example text:

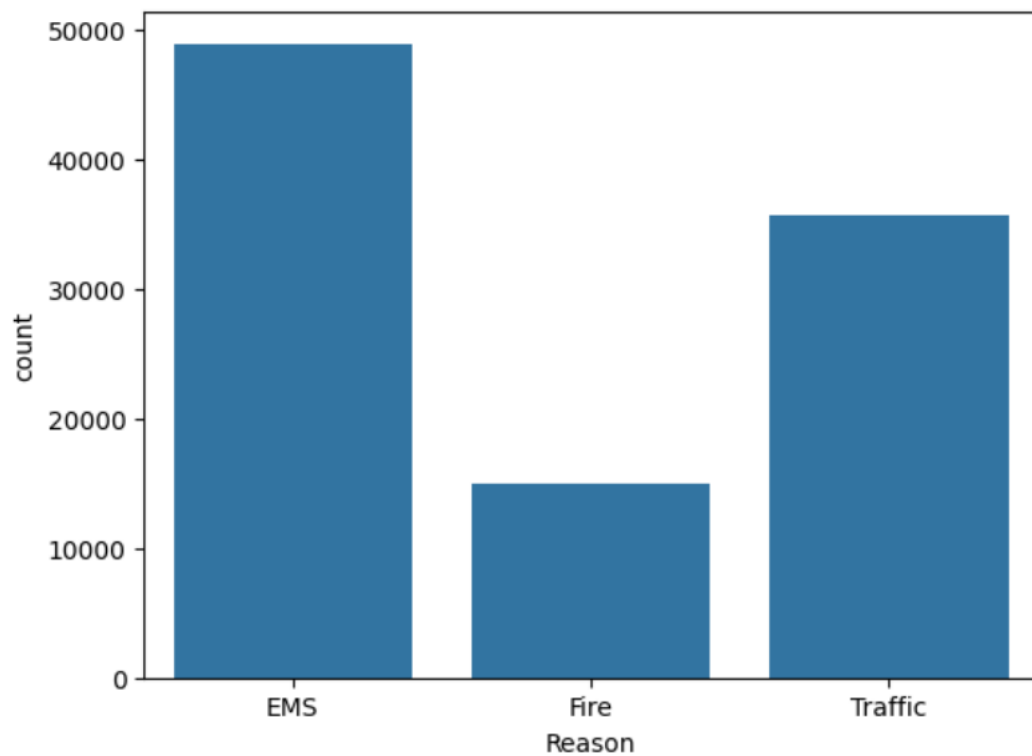
The timestamp field was converted into datetime format and additional temporal features such as hour, month, and day of week were extracted to support time-based analysis.

4. Exploratory Data Analysis

Emergency Categories

```
[11]: sns.countplot(x='Reason',data=df)
```

```
[11]: <Axes: xlabel='Reason', ylabel='count'>
```

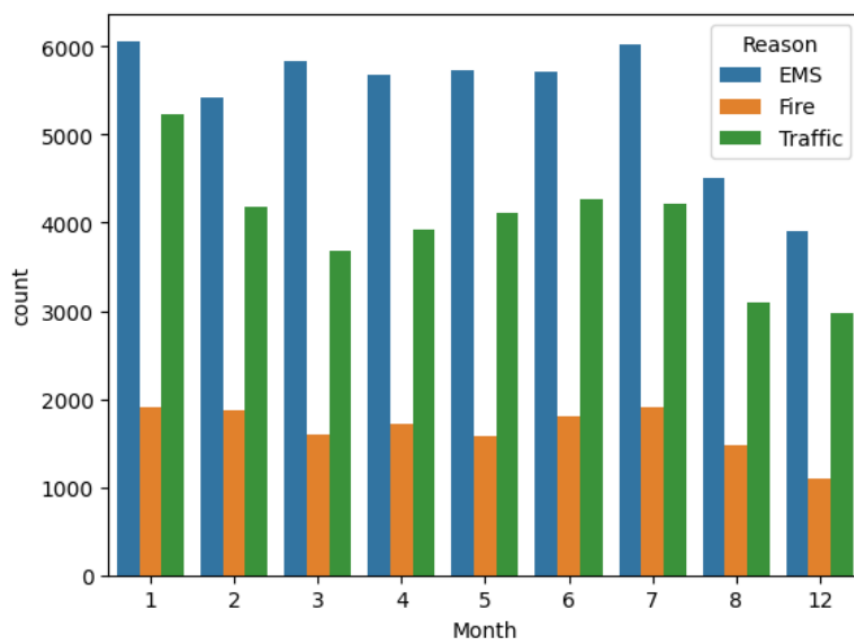


EMS calls represented the largest category of emergencies, followed by Traffic incidents and Fire-related calls.

Monthly Analysis of Emergency

```
[20]: sns.countplot(x='Month',data=df,hue='Reason')
```

```
[20]: <Axes: xlabel='Month', ylabel='count'>
```

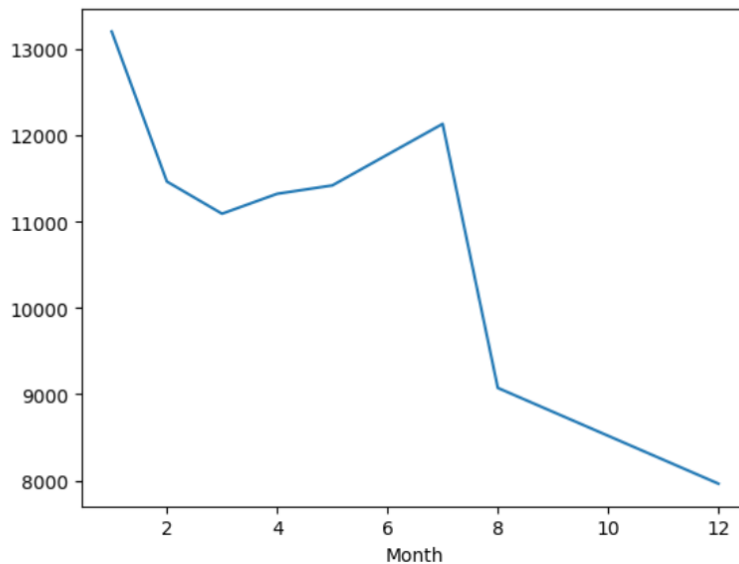


Monthly analysis showed that EMS calls dominated emergency demand throughout the year, followed by Traffic and Fire incidents. These trends can help improve resource allocation and emergency response planning.

Monthly Emergency Call Trend Analysis

```
[22]: byMonth['twp'].plot()
```

```
[22]: <Axes: xlabel='Month'>
```

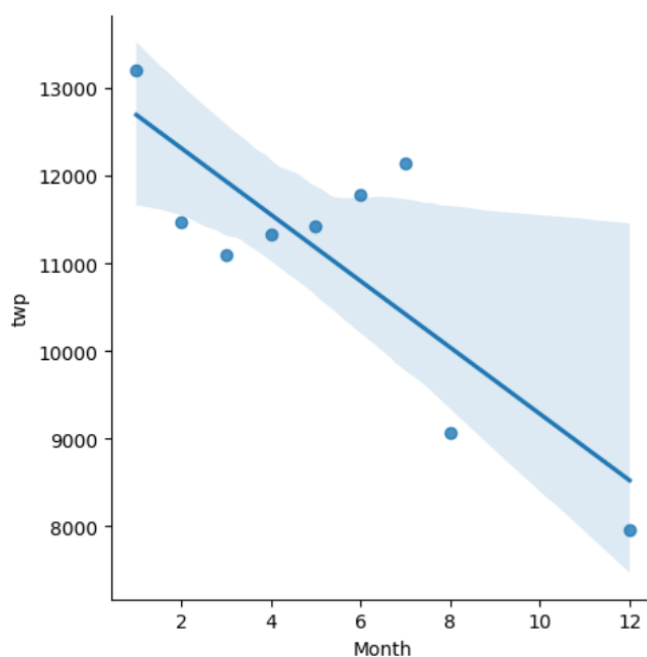


Monthly trend analysis revealed fluctuations in emergency call volumes, with higher demand during the first half of the year and lower volumes in later months, indicating seasonal and data coverage effects.

Monthly Trend Analysis

```
[23]: sns.lmplot(x='Month',y='twp',data=byMonth.reset_index())
```

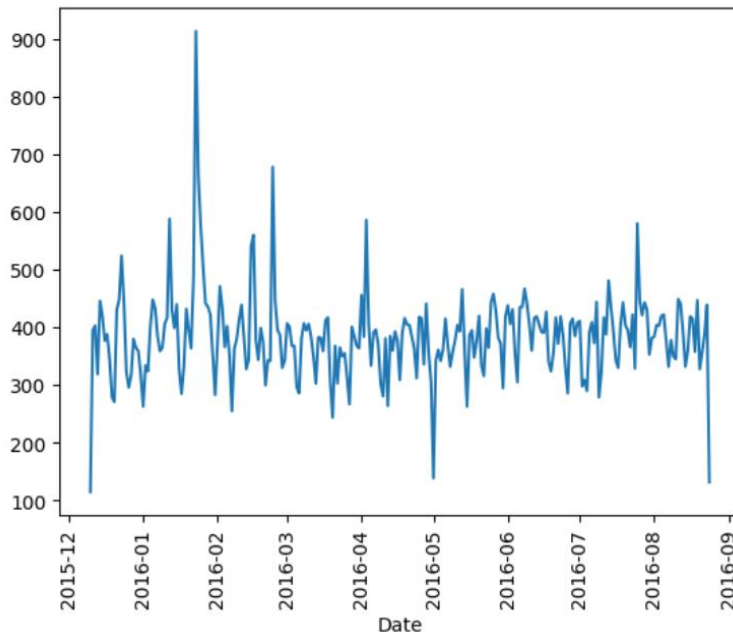
```
[23]: <seaborn.axisgrid.FacetGrid at 0x28e3ceb7170>
```



The regression trend suggests a gradual decrease in emergency call volume over time, although this pattern is likely affected by partial data availability in the final months of the dataset.

Daily Emergency Call Trend

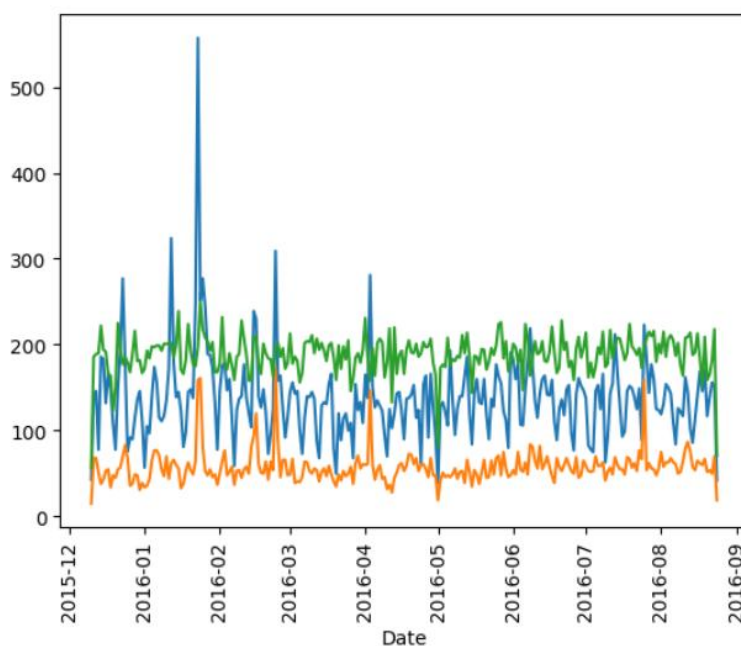
```
[25]: df.groupby('Date').count()['twp'].plot();  
plt.xticks(rotation=90);
```



Daily emergency call analysis revealed consistent call volumes with occasional spikes and drops. The maximum number of calls were between January 2016 and February 2016.

Daily Emergency Call Trends by Category

```
[43]: df[df['Reason']=='Traffic'].groupby('Date').count()['twp'].plot()  
df[df['Reason']=='Fire'].groupby('Date').count()['twp'].plot()  
df[df['Reason']=='EMS'].groupby('Date').count()['twp'].plot()  
plt.xticks(rotation=90);
```

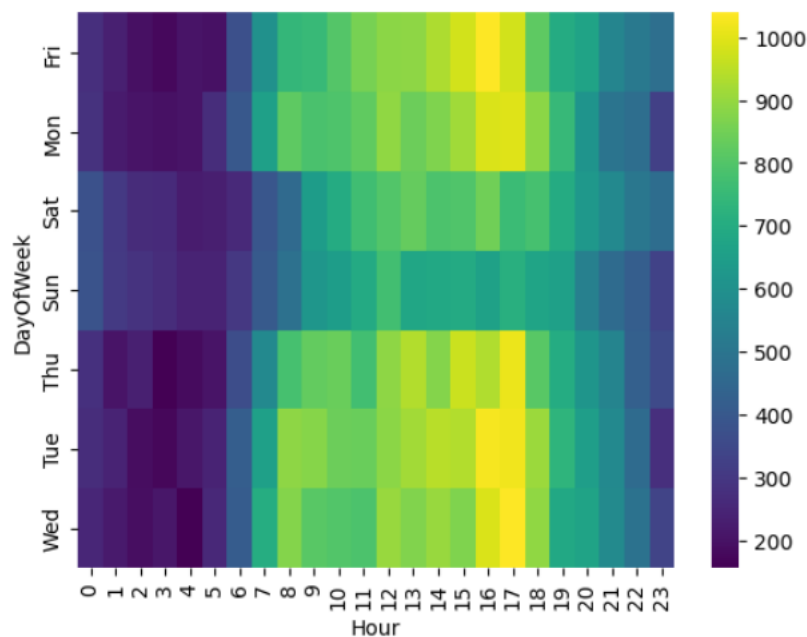


Daily category-wise analysis revealed that EMS calls consistently recorded the highest call volumes, followed by Traffic and Fire incidents. Several sudden spikes were observed, particularly in EMS and Traffic calls, indicating periods of unusually high emergency demand possibly caused by major incidents, severe weather conditions, or public events. Temporary drops in call volumes were also observed, which may be attributed to lower emergency activity or incomplete data collection on certain dates.

Hourly Distribution of 911 Calls by Day of Week using Heatmap

```
[31]: sns.heatmap(dayHour,cmap='viridis')
```

```
[31]: <Axes: xlabel='Hour', ylabel='DayOfWeek'>
```



The heatmap reveals strong temporal patterns in emergency call activity. Call volumes remain low during the early morning hours (00:00–05:00), increase rapidly after 06:00, and reach their highest levels during the afternoon period between 15:00 and 17:00. Weekdays exhibit significantly higher call frequencies than weekends, suggesting that commuting and work-related activities contribute substantially to emergency incidents.

5. Emergency Type Prediction

	precision	recall	f1-score	support
0	0.55	0.67	0.60	9734
1	0.19	0.06	0.09	2878
2	0.46	0.44	0.45	7287
accuracy			0.50	19899
macro avg	0.40	0.39	0.38	19899
weighted avg	0.46	0.50	0.47	19899

A Random Forest classifier was trained to predict the emergency category using temporal and geographic attributes. The model achieved an overall accuracy of 50%. It performed best in identifying EMS incidents but struggled to classify fire-related emergencies due to class imbalance and limited predictive features. These results suggest that additional contextual information is required for reliable emergency type prediction.

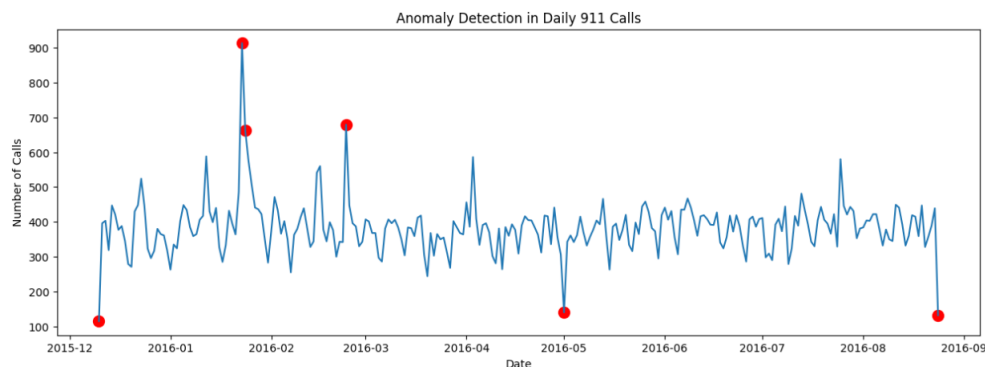
6. ZIP Code Clustering

```
[35]: cluster
      0    9405
      1   36677
      2   17331
      3   23224
      Name: total_calls, dtype: int64
```

This KMeans method is used to count the number of emergency calls from each cluster and the cluster 0 had the maximum number of calls.

7. Anomaly Detection

```
timeStamp
2015-12-10    115
2016-01-23    913
2016-01-24    662
2016-02-24    679
2016-05-01    140
2016-08-24    132
dtype: int64
```

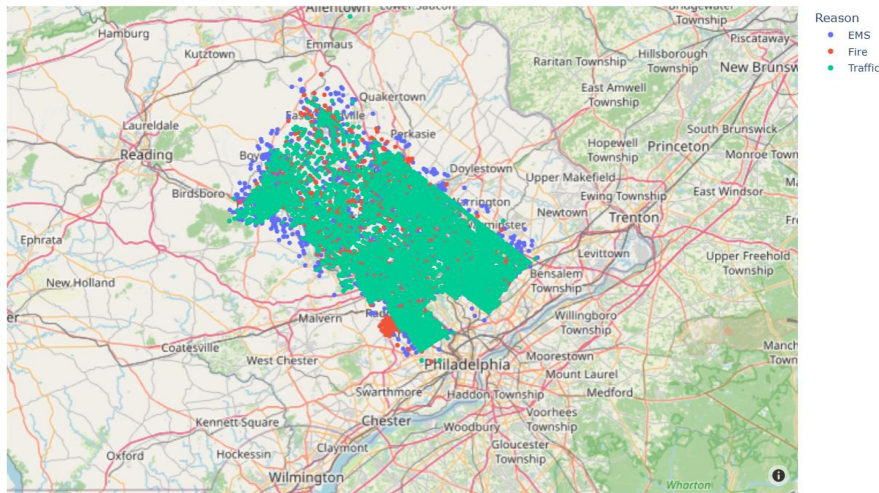


Daily 911 call volumes were analyzed to identify abnormal demand patterns. The Z-score technique was applied to measure the deviation of each day's call count from the historical average. Days with absolute Z-scores greater than 3 were flagged as anomalies. The analysis identified significant spikes in emergency calls during January and February 2016, which may correspond to severe weather events, large-scale accidents, or other extraordinary incidents. Low-value anomalies near the beginning and end of the dataset are likely due to incomplete data collection periods rather than actual decreases in emergency activity.

Why Z-score?

Z-score was used to measure how far each day's call volume deviated from the historical average in terms of standard deviations. Days with absolute Z-scores greater than three were classified as anomalies.

8. Geospatial Analysis



Geographic analysis revealed concentrated emergency hotspots within specific regions of Montgomery County. These hotspots indicate areas with consistently high emergency demand and can support strategic placement of emergency response resources.

9. Key Findings

- EMS incidents were the most frequent emergency type.
- Emergency demand peaked during afternoon hours.
- Several ZIP codes exhibited significantly higher call volumes.
- Multiple anomalous days were identified.
- Emergency incidents were geographically concentrated in a few hotspot regions.

10. Future Improvements

- Demand forecasting using time-series models.
- Real-time emergency monitoring dashboards.
- Route optimization for emergency vehicles.
- Advanced deep learning approaches for incident prediction.

11. Conclusion

This project demonstrates how data analytics, machine learning, and geospatial techniques can be combined to derive actionable insights from emergency response data. The findings can assist emergency management agencies in improving preparedness, resource allocation, and operational efficiency.